

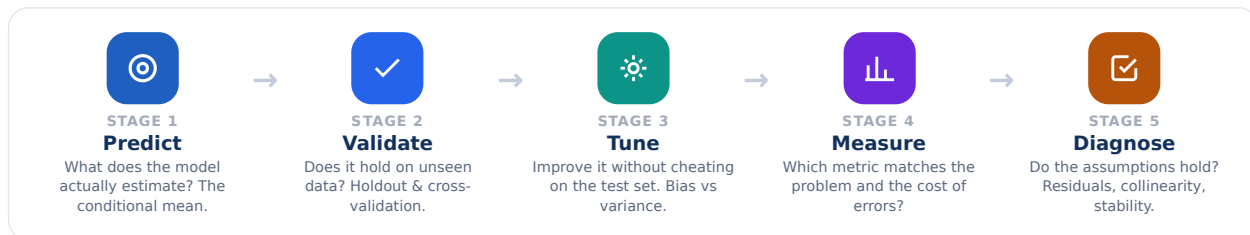
THE WHOLE JOB IN ONE LINE
 “Would this model still be right on a slab it has never seen before?”

The Evaluation Workflow

From “what does it predict?” to “can I trust it?”

Evaluation is not one number at the end — it is five questions asked in order. Skip one and you can ship a model that looks brilliant and fails in production.

THE FIVE-STAGE WORKFLOW



The one question behind it all

Every stage is a way of answering one thing honestly: **how well does this model do on data it has never seen?** Training performance is a diagnostic, not an achievement.

Fit $\neq$ Generalization Accuracy $\neq$ Validity
 Software ran $\neq$ Answer is right

Three verdicts you're trying to avoid



GOLDEN RULES OF HONEST EVALUATION

- Judge on data the model never saw.** The cross-validation score is the truth; the training score is just a diagnostic.
- Match the metric to the decision,** not to habit. The cost of each error picks the metric.
- Keep one test set untouched** until the very end — tune on cross-validation only.
- A model can predict well and still break assumptions** — read metrics *and* diagnostics.
- Prevent group leakage:** the same slab in train and test inflates every score.
- Never trust a single number** — read fit and generalization together, always.

★ KEY TAKEAWAY

Evaluation is a **sequence of honest questions**, not a final score. Ask all five, in order, every time.

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The Metric Cheat Sheet

Match the metric to the problem — and to the mistake that costs you money.

The first rule of metrics: the metric must match the problem type, what the model predicts, and which error is actually expensive. Pick before you model, not after.

DECISION RULE

Let the cost of the error — a missed defect vs. a false alarm — choose the metric.

REGRESSION · PREDICTING A NUMBER

METRIC	MEASURES	FORMULA	REACH FOR IT WHEN
MAE	Average error size, all errors equal	$\text{mean } y - \hat{y} $	You want a plain, robust “typical miss.”
MSE	Squared error — punishes big misses	$\text{mean } (y - \hat{y})^2$	Large errors are especially bad.

RMSE	MSE back in the target's own units	$\sqrt{\text{MSE}}$	You want MSE's penalty, readable units.
R²	Share of variance explained (0→1)	$1 - \text{SS}_r / \text{SS}_t$	Communicating fit to non-specialists.

CLASSIFICATION · A CLASS

Accuracy

$(\text{TP} + \text{TN}) / \text{all}$ · balanced only

Precision

$\text{TP} / (\text{TP} + \text{FP})$ · false alarms costly

Recall

$\text{TP} / (\text{TP} + \text{FN})$ · a miss is dangerous

F1

balance, imbalanced data

ROC-AUC

separability across all thresholds; threshold-free comparison

The confusion matrix

	Pred +	Pred −
Act +	TP	FN missed
Act −	FP false alarm	TN

Every classification metric is just a ratio of these four counts.

Clustering · no labels

Silhouette — cohesion vs. separation (−1→+1). No ground truth; judge how tight the groups are.

Adjusted Rand Index — agreement with known labels, corrected for chance.

Quantile · a percentile

Pinball loss — asymmetric error at quantile τ ($\tau=0.5 \rightarrow \text{MAE}$; $\tau=0.9$ penalizes under-prediction 9×).

Pseudo R² — fit vs. a null model. McFadden / Koenker-Machado form.

R² and RMSE measure the wrong target here — they assume you predicted the mean.

The costed-error rule

Missing a defect is dangerous → optimize **Recall**

False alarms waste money → optimize **Precision**

Big misses ruin you → prefer **RMSE/MSE**

Imbalanced classes → never trust **Accuracy**

★ KEY TAKEAWAY

The metric must match the **problem type** and the **cost of the mistake**. Choose it before you model.

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MODEL TUNING & EVALUATION · POSTER 3

Overfitting & Validation Decoder

Three numbers tell you whether a model learned the signal or memorized the noise.

Read the three R² values together — never one alone. The gap between them, and the spread across splits, is the whole story of generalization.

HEADLINE NUMBER

Cross-val R²

The only score that estimates performance on brand-new data.

THE THREE R² VALUES, IN ORDER OF TRUST

① Training R²

Fit on data it **has seen**. Diagnostic only — a high value alone proves nothing.

② Holdout R²

Score on a **20% slice** held back. Your first honest generalization check.

③ Cross-val R²

Averaged over **many splits** (± spread). **This is the headline number.**

THE COMPLEXITY CURVE — WHERE OVERFITTING LIVES



Training R² only ever climbs as you add complexity. **Test R²** rises, peaks at the sweet spot, then **collapses** as the model starts memorizing noise.

Tuning is the hunt for that peak — the most complexity you can afford before the test score turns down.



Healthy model

Training R ²	0.78
Holdout R ²	0.74
CV R ²	0.72 ± 0.03

Training slightly higher; all three close; small spread.



Overfit model

Training R ²	0.96
Holdout R ²	0.42
CV R ²	0.38 ± 0.10

Huge train→test drop; large, unstable spread.



Bias vs variance

High bias / underfit: low everywhere. Too simple — add signal.

High variance / overfit: great train, poor test. Too complex — simplify or regularize.

Sweet spot: lowest *total* error on unseen data.



Grouped data → GroupKFold

If the same **slab** appears in train and test, the score is a lie.

Group by `SlabID` so whole slabs stay together — then CV

answers: “*can it predict a **brand-new** slab?*”



KEY TAKEAWAY

Read the three R²s **together**. A big train→test gap is overfitting; low-everywhere is underfitting.

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